## DTC P2504:00 [SKYACTIV-D 1.5]

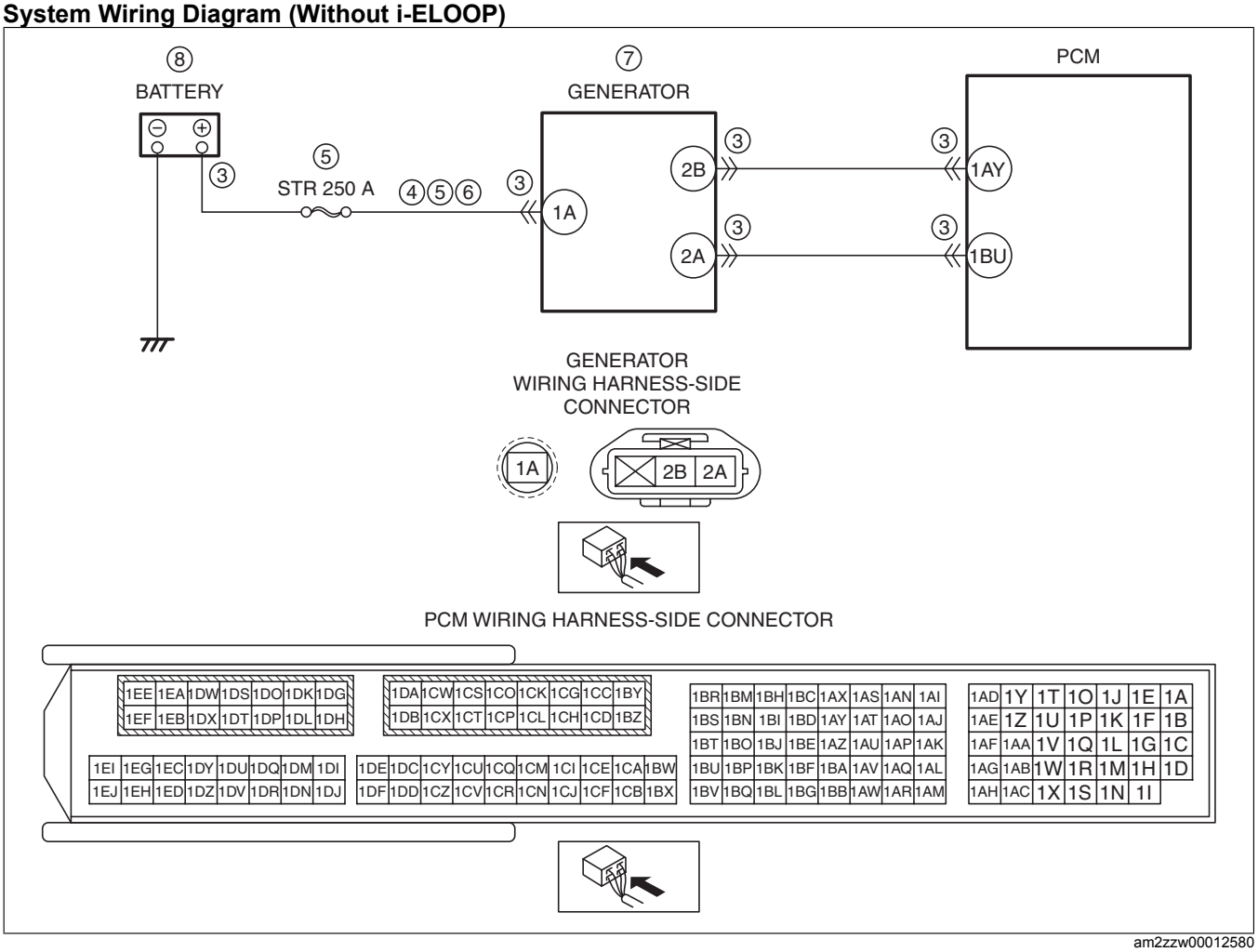
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**Note**
- To determine the malfunctioning part, proceed with the diagnostics from "Function Inspection Using M-MDS".

### Details On DTCs (Without i-ELOOP)

| | | |
|---|---|---|
| **DESCRIPTION** | **Generator system: Voltage generated by generator is high** | |
| **DETECTION CONDITION** | Determination conditions | • The voltage generated by the generator is **18.5 V or higher** or the battery voltage is **16 V or higher** for a continuous specified time. |
| | Preconditions | • While engine is running |
| | Malfunction determination period | • **5 s** period |
| | Drive cycle | • 1 |
| | Self test type | • CMDTC self test |
| | Sensor used | • PCM<br>• Generator |
| **FAIL-SAFE FUNCTION** | • Inhibits engine-stop by operating the i-stop function. | |
| **VEHICLE STATUS WHEN DTCs ARE OUTPUT** | • Flashes i-stop warning light (amber).<br>• A warning message is displayed on the center display. (With center display)<br>• Illuminates charging system warning light.<br>• The following vehicle conditions differ depending on the type of malfunction:<br>— Vehicle shock may occur due to generator load.<br>— Idling feel due to generator-stop may occur. | |
| **POSSIBLE CAUSE** | • Poor connection of the following parts:<br>— Battery<br>— Generator<br>— PCM<br>• Connector or terminal malfunction of the following parts:<br>— Battery<br>— Generator<br>— PCM<br>• Short to power supply in wiring harness between battery positive terminal and generator terminal 1A<br>• STR 250 A fuse malfunction<br>• Open circuit in wiring harness between battery positive terminal and generator terminal 1A<br>• Generator malfunction<br>• Battery malfunction<br>• PCM malfunction | |

**System Wiring Diagram (Without i-ELOOP)**




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**Function Explanation (DTC Detection Outline) (Without i-ELOOP)**

- When the charge/discharge circuit for the power supplying the vehicle is normal, the vehicle will operate normally.
- The PCM determines an over-charge malfunction by detecting that the generator terminal voltage or battery terminal voltage are abnormally high, and verification of vehicle malfunctions/safety assurance is performed.

**Repeatability Verification Procedure (Without i-ELOOP)**

1. Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-D 1.5].)
2. Start the engine.
3. Leave for **30 s** while idling with no electrical load.
4. Leave for **30 s** while idling with high electrical load.

**PID Item/Simulation Item Used In Diagnosis (Without i-ELOOP)**

- Not applicable

**Troubleshooting Diagnostic Procedure (Without i-ELOOP)**
**Intention of troubleshooting procedure**

- Step 1—6
  - — Perform an inspection of each signal transmission system.
- Step 7
  - — Perform a unit inspection of the generator.
- Step 8—10
  - — Verify that the primary malfunction is resolved and there are no other malfunctions.

| STEP | INSPECTION | RESULTS | ACTION |
|---|---|---|---|
| 1 | **PURPOSE: VERIFY RELATED SERVICE INFORMATION AVAILABILITY**<br>• Verify related Service Information availability.<br>• Is any related Service Information available? | Yes | Perform repair or diagnosis according to the available Service Information.<br>• If the vehicle is not repaired, go to the next step. |
| | | No | Go to the next step. |

| STEP | INSPECTION | RESULTS | ACTION |
|---|---|---|---|
| 2 | **PURPOSE: VERIFY IF POOR CONNECTION OF EACH PART AFFECTS DIAGNOSTIC RESULTS**<br>• Switch the ignition off.<br>• Inspect the connection condition (part installation condition, connector connection condition) for the following parts:<br>— Battery<br>— Generator<br>— PCM<br>• Is the connection condition (part installation condition, connector connection condition) for each part normal? | Yes | Go to the next step. |
| | | No | Connect each part or the connector correctly, then go to Step 8. |
| 3 | **PURPOSE: VERIFY IF CONNECTOR DAMAGE OF EACH PART AFFECTS DIAGNOSTIC RESULTS**<br>• Disconnect the connector of the following parts.<br>— Battery<br>— Generator<br>— PCM<br>• Inspect for poor connection (such as damaged/ pulled-out pins, corrosion).<br>• Is there any malfunction? | Yes | Repair or replace the connector and/or terminals, then go to Step 8. |
| | | No | Go to the next step. |
| 4 | **PURPOSE: VERIFY IF SHORT TO POWER SUPPLY IN GENERATOR CHARGE/ DISCHARGE CIRCUIT AFFECTS DIAGNOSTIC RESULTS**<br>• Verify that the battery, generator and PCM connectors are disconnected.<br>• Switch the ignition ON (engine off).<br>**Note**<br>• Another DTC may be stored by the PCM detecting an open circuit.<br>• Measure the voltage at the generator terminal 1A (wiring harness-side).<br>• Is the voltage **0 V**? | Yes | Go to the next step. |
| | | No | Refer to the wiring diagram and verify whether or not there is a common connector between battery positive terminal and generator terminal 1A.<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has a short to power supply.<br>Go to Step 8. |
| 5 | **PURPOSE: INSPECT FUSE**<br>• Switch the ignition off.<br>• Remove the STR 250 A fuse.<br>• Inspect the STR 250 A fuse.<br>• Is there any malfunction? | Yes | If the fuse is blown:<br>• Refer to the wiring diagram and verify whether or not there is a common connector between battery positive terminal and generator terminal 1A.<br>**If there is a common connector:**<br>— Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground.<br>— Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>— Repair or replace the wiring harness which has a short to ground.<br>— Replace the fuse.<br>If the fuse is damaged:<br>• Replace the fuse.<br>Go to Step 8. |
| | | No | Reinstall the STR 250 A fuse, then go to the next step. |

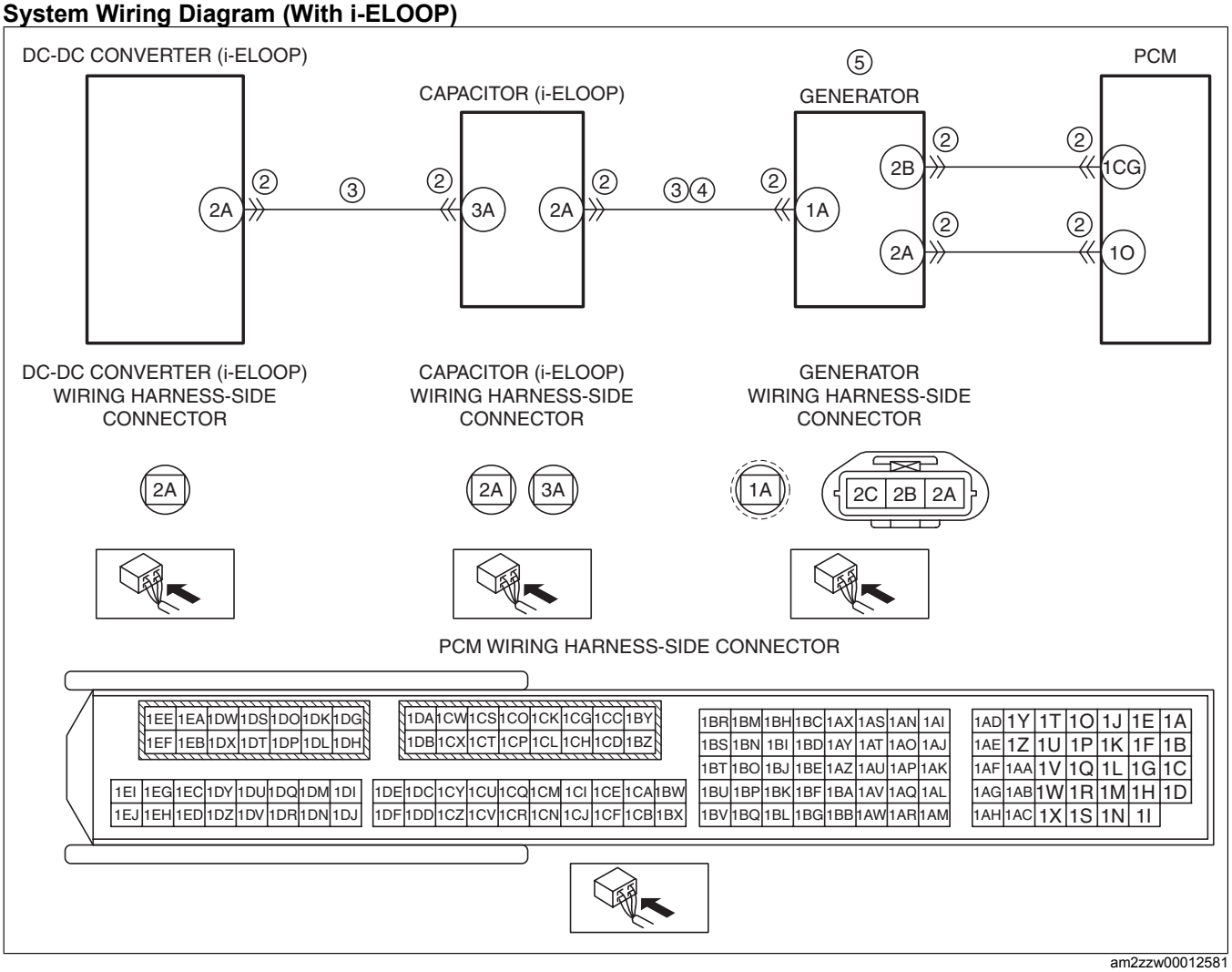
| STEP | INSPECTION | RESULTS | ACTION |
|---|---|---|---|
| 6 | **PURPOSE: VERIFY IF OPEN CIRCUIT IN GENERATOR CHARGE/DISCHARGE CIRCUIT AFFECTS DIAGNOSTIC RESULTS**<br>• Verify that the battery, generator and PCM connectors are disconnected.<br>• Inspect for continuity between battery positive terminal (wiring harness-side) and generator terminal 1A (wiring harness-side).<br>• Is there continuity? | Yes | Go to the next step. |
| | | No | Refer to the wiring diagram and verify whether or not there is a common connector between battery positive terminal and generator terminal 1A.<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has an open circuit.<br>Go to Step 8. |
| 7 | **PURPOSE: DETERMINE INTEGRITY OF GENERATOR**<br>• Inspect the generator.<br>(See GENERATOR INSPECTION [WITHOUT i-ELOOP (SKYACTIV-D 1.5)].)<br>• Is there any malfunction? | Yes | Replace the generator, then go to the next step.<br>(See GENERATOR REMOVAL/INSTALLATION [WITHOUT i-ELOOP (SKYACTIV-D 1.5)].) |
| | | No | Go to the next step. |
| 8 | **PURPOSE: VERIFY CONDITIONS OF BATTERY**<br>• Inspect the battery.<br>(See BATTERY INSPECTION.) | — | Follow the inspection instructions, then go to the next step. |
| 9 | **PURPOSE: VERIFICATION OF VEHICLE REPAIR COMPLETION**<br>• Always reconnect all disconnected connectors.<br>• Clear the DTC from the PCM memory using the M-MDS.<br>(See CLEARING DTC [SKYACTIV-D 1.5].)<br>• Implement the repeatability verification procedure.<br>(See Repeatability Verification Procedure (Without i-ELOOP).)<br>• Perform the DTC Reading Procedure.<br>(See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].)<br>• Is the same DTC present? | Yes | Repeat the inspection from Step 1.<br>• If the malfunction recurs, replace the PCM.<br>(See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)<br>Go to the next step. |
| | | No | Go to the next step. |
| 10 | **PURPOSE: VERIFY IF THERE IS ANY OTHER MALFUNCTION**<br>• Is any other DTC or pending code stored? | Yes | Go to the applicable DTC inspection.<br>(See DTC TABLE [SKYACTIV-D 1.5].) |
| | | No | DTC troubleshooting completed. |

### Details On DTCs (With i-ELOOP)

| DESCRIPTION | Generator system: Voltage generated by generator is high | |
|---|---|---|
| DETECTION CONDITION | Determination conditions | • Any one of the following conditions is met:<br>— Field current value returned by generator is abnormally larger than field current value demanded of generator, or condition in which battery voltage is **15.9 V or more** continues for specified period<br>— Capacitor (i-ELOOP) voltage is **26.8 V or more**, or condition in which over-voltage notification signal from generator continues for specified period.<br>— Condition in which overheating detection signal output from generator continues for specified period. |
| | Preconditions | • While engine is running |
| | Malfunction determination period | • **5 s** period |
| | Drive cycle | • 1 |
| | Self test type | • CMDTC self test |
| | Sensor used | • PCM<br>• Generator |
| FAIL-SAFE FUNCTION | • Inhibits engine-stop by operating the i-stop function. | |

| DESCRIPTION | Generator system: Voltage generated by generator is high |
|---|---|
| VEHICLE STATUS WHEN DTCs ARE OUTPUT | • Flashes i-stop warning light (amber).<br>• A warning message is displayed on the center display. (With center display)<br>• Illuminates charging system warning light.<br>• The following vehicle conditions differ depending on the type of malfunction.<br>— A vehicle shock may occur due to generator load.<br>— Idling feel due to generator-stop may occur. |
| POSSIBLE CAUSE | • Poor connection of the following parts:<br>— Battery<br>— PCM<br>— Generator<br>— Capacitor (i-ELOOP)<br>— DC-DC converter (i-ELOOP)<br>• Connector or terminal malfunction of the following parts:<br>— Battery<br>— PCM<br>— Generator<br>— Capacitor (i-ELOOP)<br>— DC-DC converter (i-ELOOP)<br>• Short to power supply in wiring harness between the following terminals:<br>— DC-DC converter (i-ELOOP) terminal 2A—Capacitor (i-ELOOP) terminal 3A<br>— Capacitor (i-ELOOP) terminal 2A—Generator terminal 1A<br>• Open circuit in wiring harness between capacitor (i-ELOOP) terminal 2A—Generator terminal 1A<br>• Generator malfunction<br>• Battery malfunction<br>• Capacitor (i-ELOOP) malfunction<br>• DC-DC converter (i-ELOOP) malfunction<br>• PCM malfunction |

**System Wiring Diagram (With i-ELOOP)**




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**Function Explanation (DTC Detection Outline) (With i-ELOOP)**

- If the PCM detects any of the following conditions, it determines that the generator is over-charging and verification of vehicle malfunctions/safety assurance is performed.
  - — Vehicle capacitor (i-ELOOP) and battery voltage are specified value or more
  - — Large difference between field current value demanded of generator from PCM and field current value returned from generator
  - — Signal from generator notifying of malfunction is received

**Repeatability Verification Procedure (With i-ELOOP)**

1. Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-D 1.5].)
2. Start the engine.
3. Leave for **30 s** while idling with no electrical load.
4. Leave for **30 s** while idling with high electrical load.

**PID Item/Simulation Item Used In Diagnosis (With i-ELOOP)**

- Not applicable

**Function Inspection Using M-MDS (With i-ELOOP)**

| STEP | INSPECTION | RESULTS | ACTION |
|---|---|---|---|
| 1 | **PURPOSE: VERIFY RELATED SERVICE INFORMATION AVAILABILITY**<br>• Verify related Service Information availability.<br>• Is any related Service Information available? | Yes | Perform repair or diagnosis according to the available Service Information.<br>• If the vehicle is not repaired, go to the next step. |
| | | No | Go to the next step. |

| STEP | INSPECTION | RESULTS | ACTION |
|---|---|---|---|
| 2 | **PURPOSE: VERIFY IF GENERATOR OUTPUT IS AFFECTED BY DTC OCCURRING FROM DC-DC CONVERTER (i-ELOOP) RELATED PART**<br>• Switch the ignition off, then ON (engine off).<br>• Perform the DC-DC converter (i-ELOOP) DTC inspection using the M-MDS.<br>(See DTC INSPECTION [i-ELOOP].)<br>• Are any DTCs present? | Yes | Repair or replace the malfunctioning part according to the applicable DTC troubleshooting.<br>(See DTC TABLE [i-ELOOP].)<br>Go to the next step. |
| | | No | Go to the next step. |
| 3 | **PURPOSE: VERIFY DTC**<br>• Retrieve the PCM DTCs using the M-MDS.<br>(See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].)<br>• Are any DTCs present? | Yes | Go to the applicable DTC inspection.<br>(See DTC TABLE [SKYACTIV-D 1.5].)<br>Go to the troubleshooting procedure to perform the procedure from Step 1. |
| | | No | Go to the troubleshooting procedure to perform the procedure from Step 1. |

**Troubleshooting Diagnostic Procedure (With i-ELOOP)**
**Intention of troubleshooting procedure**

- Step 1—4
  - — Perform inspection of each signal transmission system.
- Step 5
  - — Perform a unit inspection of the generator.
- Step 6—8
  - — Verify that primary malfunction is resolved and there are no other malfunctions.

| STEP | INSPECTION | RESULTS | ACTION |
|---|---|---|---|
| 1 | **PURPOSE: VERIFY IF POOR CONNECTION OF EACH PART AFFECTS DIAGNOSTIC RESULTS**<br>• Switch the ignition off.<br>• Remove the service plug.<br>(See SERVICE PLUG DISCONNECTION/ CONNECTION [i-ELOOP].)<br>• Inspect the connection condition (part installation condition, connector connection condition) for the following parts:<br>— Battery<br>— PCM<br>— Generator<br>— Capacitor (i-ELOOP)<br>— DC-DC converter (i-ELOOP)<br>• Is the connection condition (part installation condition, connector connection condition) for each part normal? | Yes | Go to the next step. |
| | | No | Connect each part or the connector correctly, then go to the next step. |
| 2 | **PURPOSE: VERIFY IF CONNECTOR DAMAGE OF EACH PART AFFECTS DIAGNOSTIC RESULTS**<br>• Disconnect the connector of the following parts.<br>— Battery<br>— PCM<br>— Generator<br>— Capacitor (i-ELOOP)<br>— DC-DC converter (i-ELOOP)<br>• Inspect for poor connection (such as damaged/ pulled-out pins, corrosion).<br>• Is there any malfunction? | Yes | Repair or replace the connector and/or terminals, then go to the next step. |
| | | No | Go to the next step. |

| STEP | INSPECTION | RESULTS | ACTION |
|---|---|---|---|
| 3 | **PURPOSE: VERIFY IF SHORT TO POWER SUPPLY IN EACH WIRING HARNESS AFFECTS DIAGNOSTIC RESULTS**<br>• Verify that the battery, PCM, generator, capacitor (i-ELOOP), and DC-DC converter (i-ELOOP) connectors are disconnected.<br>• Switch the ignition ON (engine off).<br><br>**Note**<br>• Another DTC may be stored by the PCM detecting an open circuit.<br>• Measure the voltage at the following terminals (wiring harness-side):<br>— Capacitor (i-ELOOP) terminal 3A<br>— Capacitor (i-ELOOP) terminal 2A<br>• Is the voltage **0 V**? | Yes | Go to the next step. |
| | | No | Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals:<br>• DC-DC converter (i-ELOOP) terminal 2A—Capacitor (i-ELOOP) terminal 3A<br>• Capacitor (i-ELOOP) terminal 2A—Generator terminal 1A<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has a short to power supply.<br>Go to the next step. |
| 4 | **PURPOSE: VERIFY IF OPEN CIRCUIT IN GENERATOR CHARGE/DISCHARGE CIRCUIT AFFECTS DIAGNOSTIC RESULTS**<br>• Verify that the battery, PCM, generator, capacitor (i-ELOOP), and DC-DC converter (i-ELOOP) connectors are disconnected.<br>• Switch the ignition off.<br>• Inspect for continuity between capacitor (i-ELOOP) terminal 2A (wiring harness-side) and generator terminal 1A (wiring harness-side).<br>• Is there continuity? | Yes | Go to the next step. |
| | | No | Refer to the wiring diagram and verify whether or not there is a common connector between capacitor (i-ELOOP) terminal 2A and generator terminal 1A.<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has an open circuit.<br>Go to the next step. |
| 5 | **PURPOSE: DETERMINE INTEGRITY OF GENERATOR**<br>• Inspect the generator.<br>(See GENERATOR INSPECTION [WITH i-ELOOP (SKYACTIV-D 1.5)].)<br>• Is there any malfunction? | Yes | Replace the generator, then go to the next step.<br>(See GENERATOR REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-D 1.5)].) |
| | | No | Go to the next step. |
| 6 | **PURPOSE: VERIFY CONDITIONS OF BATTERY**<br>• Inspect the battery.<br>(See BATTERY INSPECTION.) | — | Follow the inspection instructions, then go to the next step. |
| 7 | **PURPOSE: VERIFICATION OF VEHICLE REPAIR COMPLETION**<br>• Always reconnect all disconnected connectors.<br>• Reinstall the service plug.<br>(See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].)<br>• Clear the DTC from the PCM memory using the M-MDS.<br>(See CLEARING DTC [SKYACTIV-D 1.5].)<br>• Implement the repeatability verification procedure.<br>(See Repeatability Verification Procedure (With i-ELOOP).)<br>• Perform the DTC Reading Procedure.<br>(See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].)<br>• Is the same DTC present? | Yes | Repeat the inspection from Step 1.<br>• If the malfunction recurs, replace the PCM.<br>(See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)<br>Go to the next step. |
| | | No | Go to the next step. |
| 8 | **PURPOSE: VERIFY IF THERE IS ANY OTHER MALFUNCTION**<br>• Is any other DTC or pending code stored? | Yes | Go to the applicable DTC inspection.<br>(See DTC TABLE [SKYACTIV-D 1.5].) |
| | | No | DTC troubleshooting completed. |